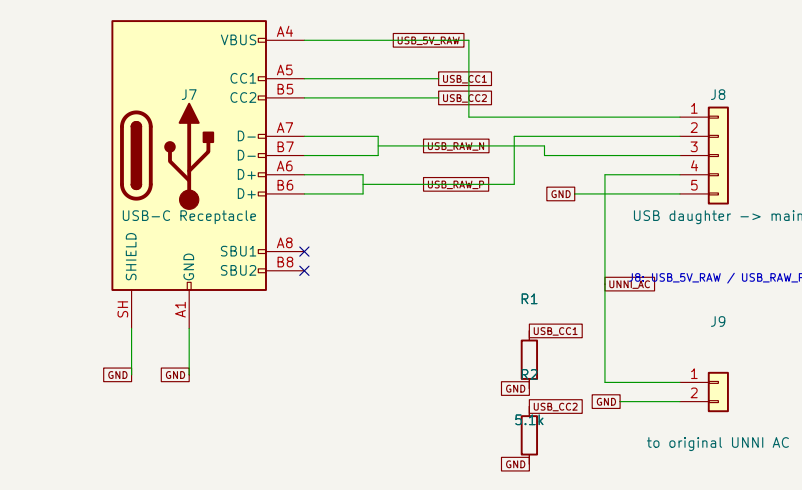


UNNI CO2 SMARTIFICATION – REV A / ESP32–C6–MINI–1

Flat schematic intentionally used for Rev A validation. Functional blocks are separated visually; all inter–block nets use named global labels.

REPLACEMENT USB–C DAUGHTERBOARD

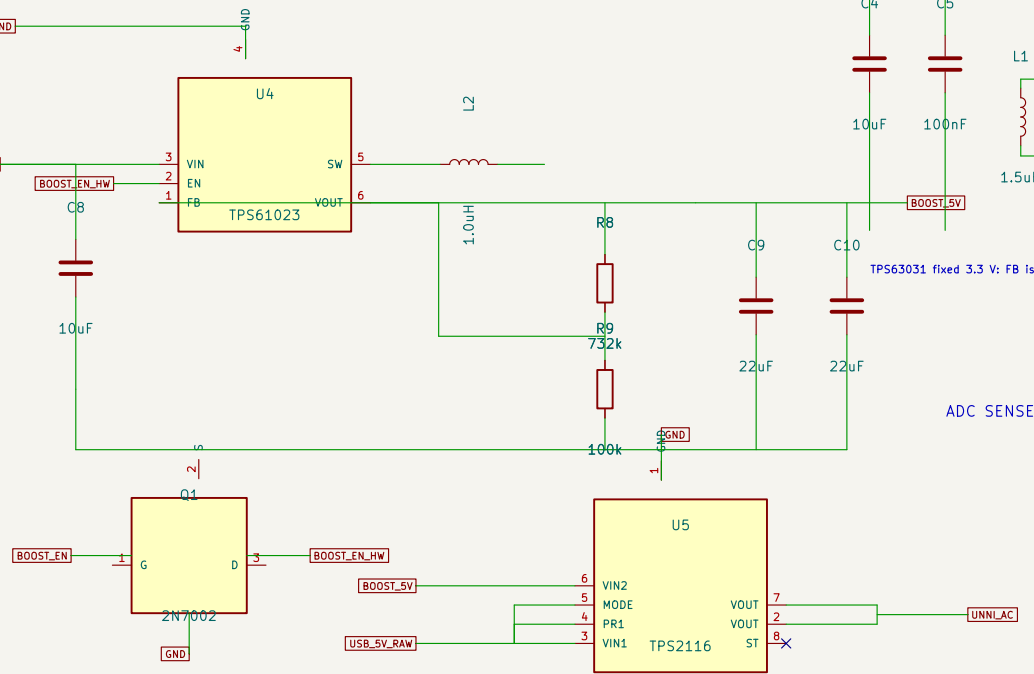


Replacement daughterboard keeps the original case USB–C opening; smartification board returns muxed UNNLAC here.

J8 inter–board pin plan: 1 USB_SV_RAW, 2 USB_RAW_P, 3 USB_RAW_N, 4 UNNLAC, 5 GND.

J9 original–Unni connection: UNNLAC + GND. Main–board J2: VBAT/UNNLDC + GND.

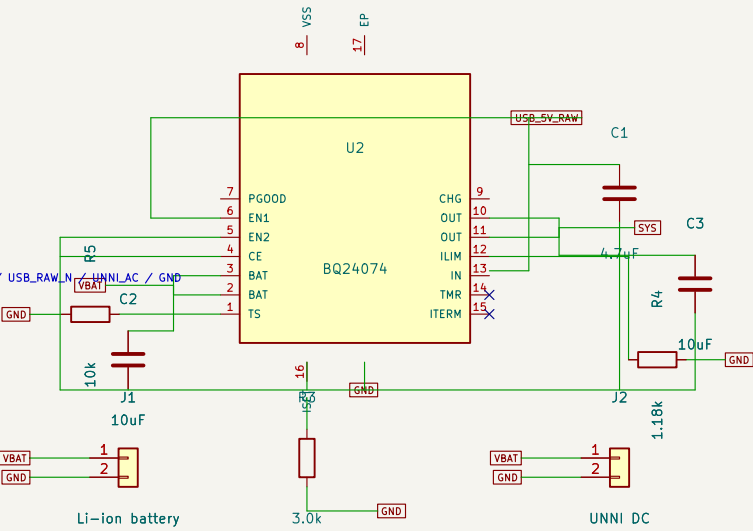
FORCED–AWAKE 5 V + USB PRIORITY MUX



Q1 / BOOST_EN_HW is the hardware–inhibit stage; final polarity/gate conditioning will be checked in ERC/design review.

TPS2116: VIN1 USB_SV_RAW has priority over VIN2 BOOST_5V; both VOUT pins are visibly tied to UNNLAC.

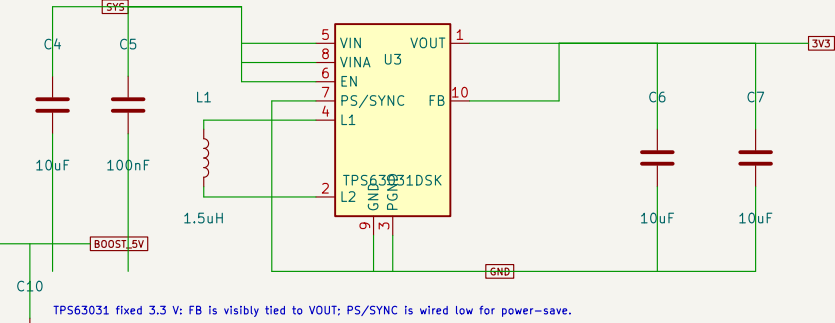
USB / LI–ION CHARGER + POWERPATH – BQ24074



Li–ion battery

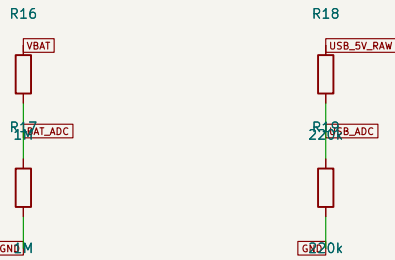
VBAT is the battery node and is also routed directly to UNNLDC.

3.3 V ESP SUPPLY – TPS63031 BUCK–BOOST

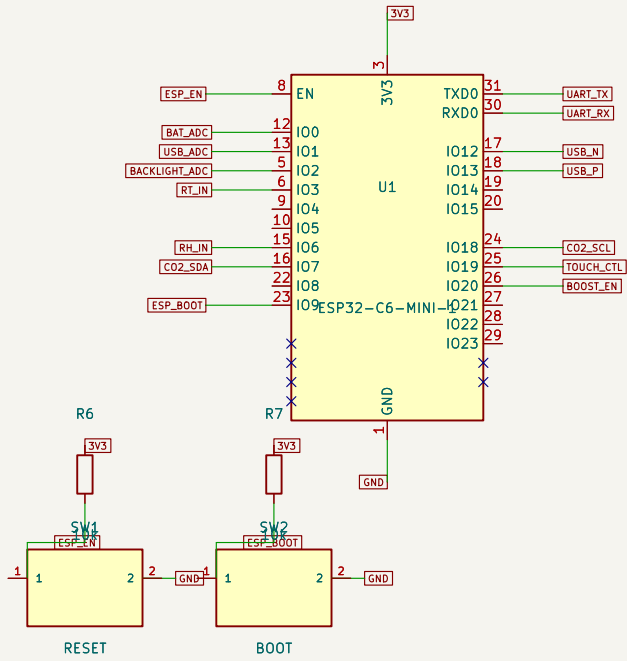


TPS63031 fixed 3.3 V; FB is visibly tied to VOUT; PS/SYNC is wired low for power–save.

ADC SENSE

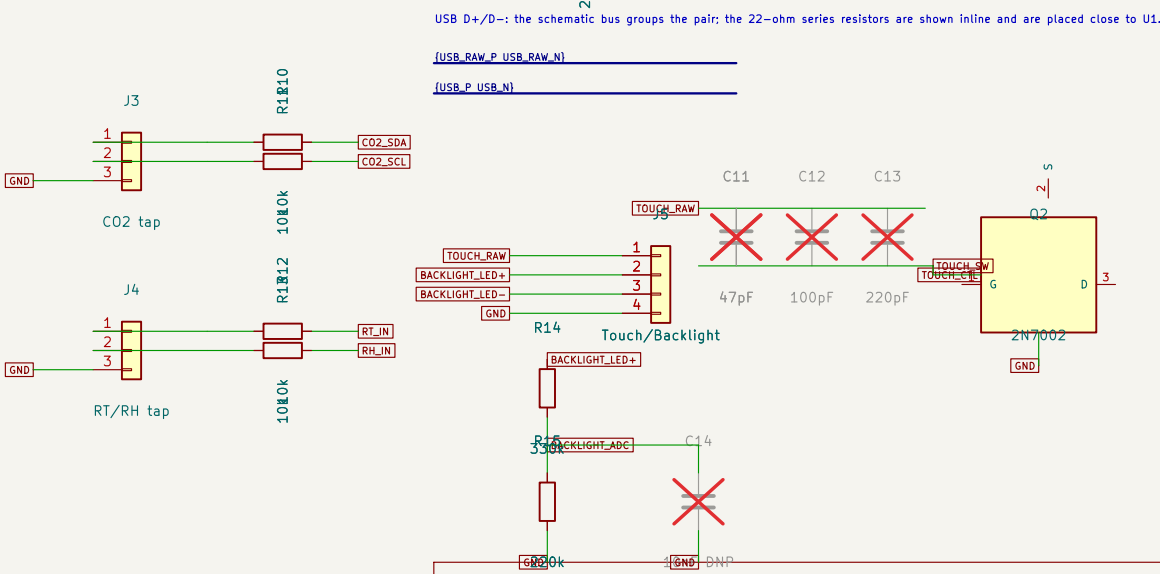


ESP32–C6–MINI–1 + NATIVE USB



GPIO4/5/8/9/15 are strapping pins; Rev A avoids them for Unni runtime signals except GPIO9 used only as BOOT.

UNNI PASSIVE SIGNAL TAPS + TOUCH/BACKLIGHT



All Unni signal connections are T–taps. Removing the smartification PCB leaves the original Unni signal paths untouched.